CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · FORMULA BANK

Chapter 21: Trigonometrical Identities

Every formula and result of the chapter on one sheet

1	Identity 1	$\sin^2 A + \cos^2 A = 1$
2	Identity 2	$1 + \tan^2 A = \sec^2 A$
3	Identity 3	$1 + \cot^2 A = \operatorname{cosec}^2 A$
4	Reciprocal relations	$\operatorname{cosec} A = \frac{1}{\sin A}, \sec A = \frac{1}{\cos A}, \cot A = \frac{1}{\tan A}$
5	Quotient relations	$\tan A = \frac{\sin A}{\cos A}, \cot A = \frac{\cos A}{\sin A}$
6	Factor forms	$(\sec A + \tan A)(\sec A - \tan A) = 1; (\operatorname{cosec} A + \cot A)(\operatorname{cosec} A - \cot A) = 1$
7	sin and cos of complementary angles	$\sin(90^\circ - A) = \cos A; \cos(90^\circ - A) = \sin A$
8	tan and cot of complementary angles	$\tan(90^\circ - A) = \cot A; \cot(90^\circ - A) = \tan A$
9	sec and cosec of complementary angles	$\sec(90^\circ - A) = \operatorname{cosec} A; \operatorname{cosec}(90^\circ - A) = \sec A$
10	Useful result	$\tan A + \cot A = \sec A \operatorname{cosec} A$
11	Useful result	$(\sin A + \cos A)^2 = 1 + 2 \sin A \cos A$
12	Degrees and minutes	$1^\circ = 60'$